

SELF-MONITORING BLOOD GLUCOSE (SMBG) DEVICE DEVELOPMENT OF A LOW-COST INVASIVE BLOOD GLUCOSE MONITORING DEVICE



- **Objective:**

Develop a low-cost, portable and user-friendly invasive Self-Monitoring Blood Glucose (SMBG) device for measuring blood glucose concentration using an electrochemical sensing technique.

- **Solution:**

Developed an academic prototype using a disposable glucose test strip, electrochemical sensing, transimpedance amplification, Arduino UNO, signal conditioning and OLED display for processing and displaying the glucose measurement.

- **Impact:**

Provides a compact and economical approach for blood glucose monitoring and demonstrates the practical application of electrochemical biosensing, embedded systems and signal processing.

Engineering Design Process

- **Requirement & Planning:**
Identified the requirements for an invasive SMBG device, including glucose sensing, signal conditioning, digital processing, display and portable power supply.
- **System Development:**
Developed the electrochemical sensing, transimpedance amplifier, ADC processing, calibration and display modules.
- **System Integration:**
Integrated the glucose test strip, LM358/LM324 signal-conditioning circuit, Arduino UNO, OLED display, battery and charging circuit.
- **Testing & Validation:**
Tested the prototype for glucose measurement response, signal stability, display operation and overall system functionality.
- **Documentation:**
Prepared the literature review, methodology, component specifications, working principle, cost estimation, results and project documentation.

Process Table

Phase	Engineering Steps	Key Activities & Outcome
Definition	Requirement Analysis, Concept Design	Defined the need for a portable invasive SMBG device. Selected electrochemical glucose sensing as the measurement principle.
Development	System Design, Circuit Development, Integration	Developed glucose sensing, TIA, signal conditioning, ADC processing and OLED display modules.
Realization	Prototype, Testing, Validation	Built the SMBG prototype and evaluated its sensing response, signal processing and display functionality

List of Activities

- **A. Literature Survey:** Studied finger-prick glucose meters, electrochemical biosensors, GOx/GDH sensing, CGM technologies and glucose-monitoring accuracy.
- **B. Requirement Analysis:** Identified requirements for sensing, amplification, processing, display, power supply and portability.
- **C. Sensor & Strip Selection:** Selected a disposable glucose test strip based on electrochemical enzymatic sensing.
- **D. Signal Conditioning:** Designed the transimpedance amplifier and filtering circuit to convert the sensor current into a measurable voltage.
- **E. Microcontroller Development:** Implemented ADC acquisition, calibration and glucose-concentration processing using Arduino UNO.
- **F. System Integration:** Integrated the sensing circuit, Arduino UNO, OLED display, push button, battery and charging module.
- **G. Testing & Validation:** Evaluated the prototype for signal response, measurement operation, stability and overall functionality.



Activity Duration Table

ID	Activity Description	Duration (Hours)
A	Literature Survey & Requirement Analysis	30
B	Concept Design	25
C	Component Selection & Circuit Design	40
D	Prototype Development & Programming	55
E	System Integration	30
F	Testing, Calibration & Optimization	40
G	Documentation & Presentation	30
Total	Project Duration	250 Hours

Activity Precedence Table

ID	Activity	Predecessor
A	Requirement Analysis & Literature Survey	—
B	Concept Design	A
C	Component Selection & Circuit Design	B
D	Prototype Development & Programming	C
E	System Integration	D
F	Testing & Calibration	E
G	Documentation & Presentation	F

Glossary:

- **SMBG:** Self-Monitoring Blood Glucose system used for measuring blood glucose by the user.
- **GOx:** Glucose Oxidase enzyme used in electrochemical glucose sensing.
- **GDH:** Glucose Dehydrogenase enzyme used as an alternative glucose-sensing enzyme.
- **TIA:** Transimpedance Amplifier used to convert sensor current into voltage.
- **ADC:** Analog-to-Digital Converter used to convert the conditioned sensor signal into digital data.
- **MARD:** Mean Absolute Relative Difference used to assess glucose measurement accuracy.
- **ISO 15197:** Standard associated with performance requirements for blood-glucose monitoring systems.
- **OLED:** Organic Light-Emitting Diode display used to show the measured glucose value.

Core Technologies & Principles

- **Electrochemical Biosensing:**
Measures the electrical response generated by the enzymatic reaction of glucose on the test strip.
- **Enzymatic Reaction:**
GOx or GDH reacts with glucose and produces an electrochemical signal related to glucose concentration.
- **Transimpedance Amplification:**
Converts the small sensor current into a voltage suitable for microcontroller measurement.
- **ADC Processing:**
Arduino UNO converts the analog conditioned signal into digital data.
- **Calibration:**
Relates the measured electrical signal to glucose concentration.
- **Digital Display:**
OLED displays the processed glucose measurement to the user.

Concept Rules & Laws

- **Electrochemical Sensing Principle:**

The glucose concentration is related to the electrical current generated by the enzymatic reaction at the test strip.

- **Current-to-Voltage Conversion:**

The transimpedance amplifier converts the electrochemical sensor current into a measurable voltage.

- **Analog-to-Digital Conversion:**

The Arduino UNO ADC converts the analog voltage into digital data.

- **Calibration Principle:**

The measured electrical response is mapped to glucose concentration using calibration parameters.

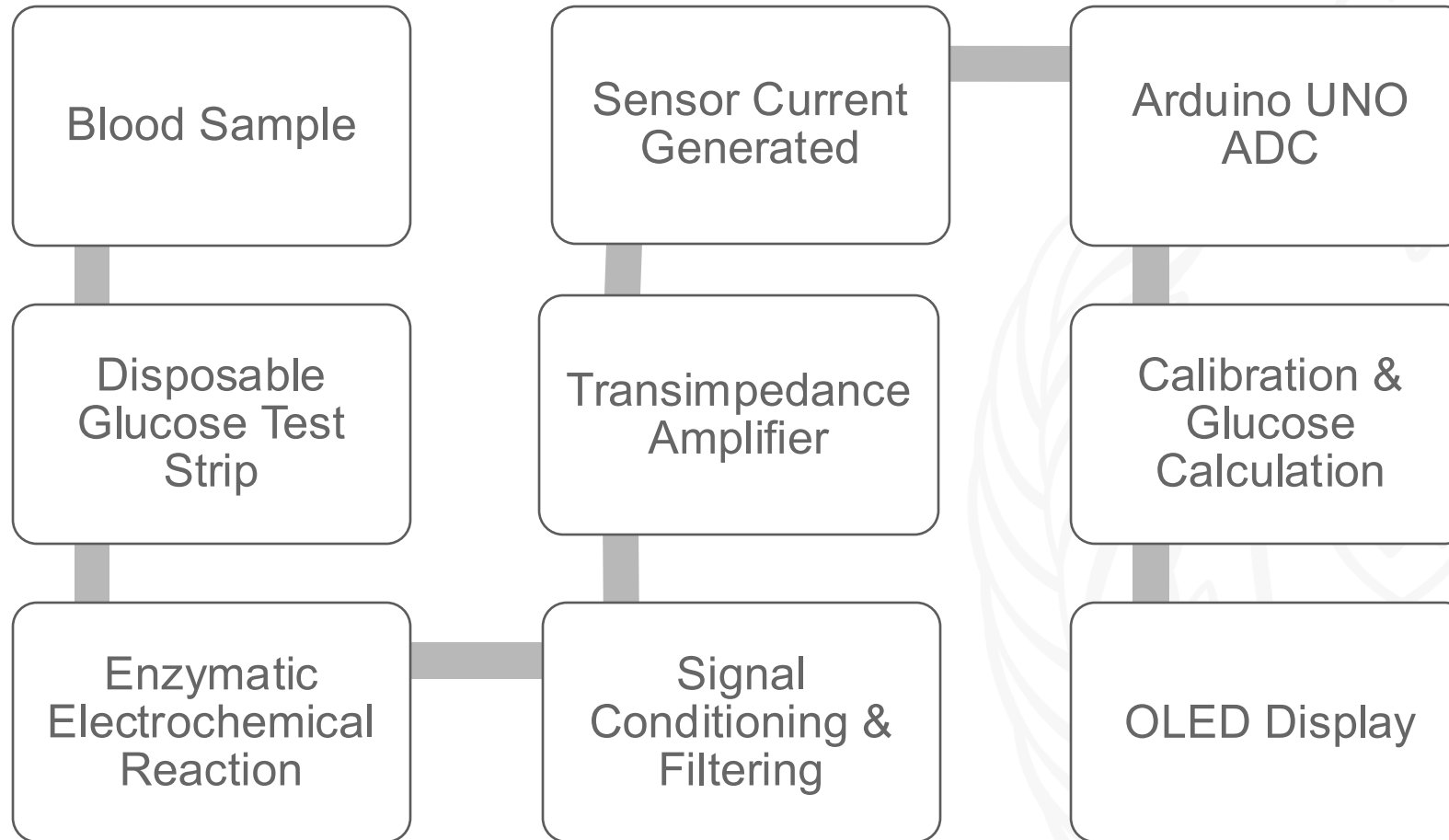
- **Accuracy Evaluation:**

Performance can be assessed using measures such as MARD and error-grid analysis, together with relevant glucose-meter accuracy standards.

System Components & Technologies

Technology	Purpose
Arduino UNO – ATmega328P	Signal acquisition, processing and control
Glucose Test Strip	Electrochemical glucose sensing
LM358 / LM324	Facial landmark detection
Precision Resistors	Gain and circuit parameter setting
Capacitors	Filtering and noise reduction
OLED 128×64 I ² C	Displays glucose measurement
Boost Converter	Voltage regulation/boosting

System Model



What-if-Analysis

Scenario	Expected System Response
No Test Strip/Sample	System waits for a valid sensing input.
Blood Sample Applied	Electrochemical reaction generates a current proportional to glucose concentration.
Weak Sensor Signal	Signal-conditioning stage amplifies the low-level current signal for ADC measurement.
Noise in Sensor Signal	Filtering and signal conditioning reduce unwanted electrical fluctuations.
Valid Glucose Measurement	Arduino processes the ADC value and displays the glucose concentration on the OLED
Low Battery	Portable power system requires charging through the TP4056 module.

Conclusion

- The Self-Monitoring Blood Glucose (SMBG) Device successfully demonstrates the development of a low-cost invasive glucose-monitoring prototype based on electrochemical enzymatic sensing. The system combines a disposable glucose test strip, transimpedance signal conditioning, Arduino UNO-based ADC processing and an OLED display to obtain and present glucose measurements. The project demonstrates the integration of biosensing, analog electronics, embedded systems and calibration techniques in a compact prototype. Further development can focus on improving measurement accuracy, calibration, noise reduction, miniaturization, battery optimization and validation against established glucose-monitoring standards.
- The PPT template itself emphasizes a complete engineering workflow from requirement definition through development, integration, testing and validation, which this SMBG content has been mapped onto.